

# A Language-Enhanced Supervision-Reduced Analysis Framework for Multibranching Vessels

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**Abstract**—It is essential to quantitatively analyze coronary arterial angiography (CAG) images for coronary interventional procedures, as they are the gold reference standard for assessing stenosis. However, mainstream methods cannot extract different vascular branches precisely due to inadequate annotated data by experts and difficulty in uniformly defining vascular branches. To address the issue, this article presents a novel framework to analyze multibranching vessels with simple annotation, which localizes the endpoints of the vascular branches first, and then extracts corresponding vessel segments based on the endpoints. Considering it is challenging for current image-based methods to distinguish sparse vessel keypoints, we propose a cross-modal language-enhanced endpoint detection (LEED) network to achieve better performance by providing linguistic knowledge obtained from clinicians. A medical term corpus is established

to convert conversation contents to standard texts. Furthermore, we devise a twin-stream vessel segmentation (TS-Seg) algorithm to segment various vascular branches based on gray-level features and detected endpoints. Besides, clinicians are allowed to manually add new endpoints and apply the TS-Seg algorithm alone to analyze any variant vascular branch that has not been defined before. Extensive experiments on two in-house and one public datasets consistently demonstrate the effectiveness and superiority of the proposed framework. Moreover, comprehensive vessel analysis reports are provided based on clinical prior knowledge to assist in preoperative surgery planning and postoperative treatment assessment, showing great potential to improve the efficiency and accuracy of clinical workflow.

**Impact Statement**—In vascular interventional surgeries, comprehensive quantitative analysis of multibranching vessels in CAG images will help clinicians make optimal surgical plans. However, it is challenging to define the vascular branches uniformly due to significant individual differences, and the tortuous vascular structures result in a heavy annotation workload, both making it difficult for existing methods to achieve precise vessel analysis. Our framework overcomes the challenges by necessitating only simple point annotations to realize the extraction of any specified vascular branch based on its two endpoints. Furthermore, it enhances the accuracy of point recognition by 3% through the integration of linguistic information obtained from clinicians. Additionally, detailed vessel analysis reports are provided to clinicians for making surgical plans and assessing treatment outcomes. This framework offers a new solution to annotation challenges in vessel segmentation and demonstrates the potential for application in intelligent personalized diagnosis and treatment for vascular diseases.

**Index Terms**—Keypoint detection, multibranching vessel analysis, supervision-reduced segmentation, vision-language model.

## I. INTRODUCTION

VASCULAR diseases, especially cardiovascular diseases, have been the world's leading cause of death for years and are still increasing worldwide [1]. Possessing the advantages of smaller surgical incisions and shorter recovery time, vascular interventional procedures are now the most popular treatment for vascular diseases [2]. Coronary arterial angiography (CAG) imaging is the gold standard for coronary interventional procedures because it can reveal various pathologies and assess the blood flow changes caused by lesions [3]. Currently, the assessment of CAG images is mostly qualitative, while quantitative analysis is meaningful for assessing the stenosis diameter objectively [4], [5], [6]. Achieving such quantitative analysis necessitates precise segmentation of every vascular branch.

Received 30 September 2025; revised 3 December 2025, 22 January 2026, and 2 March 2026; accepted 1 April 2026. Date of publication 7 April 2026; date of current version 31 August 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFF1206902 and Grant 2024YFE0215200; in part by the National Natural Science Foundation of China under Grant 62303463, Grant 62073325, Grant 62003343, Grant 62222316, and Grant U20A20224; in part by the Beijing Natural Science Foundation under Grant L232137, Grant L246047, Grant L258031, and Grant M22008; and in part by the Youth Innovation Promotion Association of CAS under Grant 2020140. This article was recommended for publication by Associate Editor Deepika Koundal upon evaluation of the reviewers' comments. (*Corresponding authors: Shi-Qi Liu; Zeng-Guang Hou.*)

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Digital Object Identifier 10.1109/TAI.2026.3681535

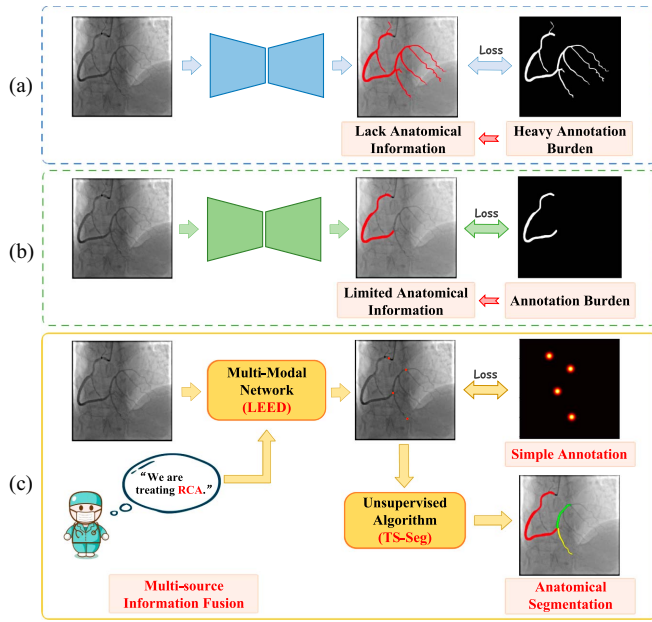


Fig. 1. Difference among vessel segmentation methods. (a) Vessel segmentation methods that differentiate between the vascular foreground and background. (b) Vessel segmentation methods that focus on the main vessel branch. (c) Vessel anatomical segmentation method proposed by us.

Large datasets with high-quality annotations are essential for applying artificial intelligence (AI) technologies to help with the analysis of medical images [7]. However, due to the significant individual variability in coronary vascular branches [8], it is difficult to uniformly define those small branching vessels. As a result, branch-wise annotation can be time-consuming and exhausting even for professional doctors, making it challenging to apply mainstream deep-learning methods to achieve multi-branching vessel semantic segmentation. Some researchers who were not medical professionals conducted a small amount of annotation themselves [9], which only differentiated between the vascular foreground and background as shown in Fig. 1(a), without delving into a detailed vascular anatomical structure, i.e., the different branches of a vascular tree. Nevertheless, vascular anatomical structure is crucial for making surgical plans because lesions in different vessel branches could lead to different intervention paths. With the advancement of interdisciplinary collaboration between medicine and engineering, some researchers have gained access to expertise and collected main vessel segment labels annotated by cardiologists [10], [11], [12], including the left anterior descending (LAD), left circumflex (LCX), and right coronary artery (RCA) as illustrated in Fig. 1(b). However, multi-branching vessels, including the main vessel segment and other branching vessels, jointly influence clinicians' decisions in surgical planning [6]. Hence, it has limited clinical application to analyze only the main vessels. To date, no existing approach has been able to achieve accurate semantic segmentation of small vascular branches yet.

Compared to natural images, medical images suffer from multiple challenges such as low signal-to-noise ratio (SNR) and isolated data sources. Therefore, fully leveraging the available information is beneficial for achieving better performance. In surgical settings, clinicians frequently communicate with

each other, and such conversations often contain crucial domain knowledge. Specifically, in vascular interventional procedures, language information may indicate the angiographic position of CAG images or specify the target vessel being treated. The former provides context about vascular morphology from a particular perspective, while the latter directs attention to the local region of interest. By incorporating these clinically relevant language cues, models can better focus on meaningful features and thereby enhance overall performance. Moreover, compared to integrating other image modalities, leveraging language information is more clinically feasible as it would not require extra examinations for the patients. However, there has been no research on the cross-modal fusion of language and CAG images before, making its implementation a challenge.

To address the issues mentioned above, this article proposes a novel language-enhanced supervision-reduced analysis framework for multi-branching vessels. The schematic is illustrated in Fig. 1(c). To solve the problem of the inability to uniformly define multi-branching vessels and the heavy annotation burden, this article proposes a flexible strategy to achieve vascular branch semantic segmentation in an adjustable two-stage manner. First, a language-enhanced endpoint detection (LEED) network is devised to automatically localize the endpoints of the main vascular branches. Second, a designed unsupervised twin-stream vessel segmentation (TS-Seg) algorithm will generate vascular branch segmentation results based on the endpoint locations. For better keypoint detection performance, relevant medical linguistic information is extracted and standardized as text embeddings based on an established medical term corpus. Then, they are utilized for feature alignment at different levels in the LEED network. To ensure the segmentation accuracy, we have devised several techniques in the TS-Seg algorithm, including the inner noise filter (INF) for preventing growth blockage, the search-based vessel pruning (SVP) for removing excess branches, and the contrastive selection algorithm for refining edge pixels. With this two-stage structure, for one thing, our framework enables vascular branch semantic segmentation by only requiring point-level annotations for training the LEED, reaching the goal of reducing supervision conditions. For another, besides analyzing main vascular branches, it allows manual intervention to flexibly add new endpoints prior to the TS-Seg algorithm, making it feasible to segment undefined vascular branches. In this way, it can adapt to complex clinical scenarios with variant vascular structures. Finally, integrating clinical prior knowledge and extracted morphological information, a detailed vessel analysis report is generated for preoperative surgery planning and postoperative treatment assessment, which is promising to be directly applied in clinical surgeries.

The main contributions can be summarized as follows.

- 1) To the best of our knowledge, this is the first cross-modal approach named LEED to leverage linguistic information for better vessel keypoint detection. A medical term corpus is established to convert conversation contents to standard texts, which not only adapt to different clinical expressions but also are helpful in promoting cross-modal fusion.
- 2) A novel supervision-reduced and flexible strategy is proposed to realize vascular branch semantic segmentation

without the requirement of complex annotations. Based on the endpoints detected by LEED, the presented TS-Seg algorithm can extract various vascular branches accurately. Besides, clinicians are allowed to add the endpoints manually to analyze any variant vascular branch that has not been defined before.

- 3) Extensive experiments are conducted on two in-house CAG datasets with more than 500 cases and a public dataset, validating the model's state-of-the-art performance. Furthermore, comprehensive and accurate vessel analysis reports are generated based on clinical prior knowledge, showing great potential to assist in clinical diagnosis and surgical planning.

The rest of the article is organized as follows. An overview of the existing methods related to keypoint detection and vessel segmentation is introduced in Section II. The details of the proposed framework are presented in Section III, followed by the corresponding experiment results in Section IV. Finally, Section V summarizes the whole article.

## II. RELATED WORK

### A. Keypoint Detection

Most of the current keypoint detection methods are traditional vision-based models. Early works such as DeepPose [13] realize keypoint detection by directly regressing the coordinates of the points. Later, it is believed that mapping from the input image to XY location could weaken generalization ability, so mainstream methods predict Gaussian heatmaps representing possibilities of being keypoints [14], [15], [16], [17], [18], [19], [20], [21]. Newell et al. [14] proposed a symmetric Stacked Hourglass network to capture information across multiple scales, and the authors [15], [16] further extended this structure for more complicated tasks. Sun et al. [17] designed a novel HRNet, consisting of parallel high-to-low resolution subnetworks, to capture rich high-resolution representations. Based on this work, Cheng et al. [18] proposed HigherHRNet to make the model robust to scale variation. Li et al. [20] explicitly embedded keypoints as extra tokens to learn the relationship among different points. Xu et al. [21] proposed a simple but flexible model ViTPose based on plain transformers.

In recent years, there has emerged a series of works exploring the vision-language interaction, among which large pretrained vision-language models such as CLIP [22] have shown great success on a wide range of downstream computer vision tasks, such as segmentation [23], [24] and pose estimation [25], [26]. Zhang et al. [25] integrated language knowledge into animal pose estimation at both spatial and feature levels. Besides the semantic information of the joints, Hu et al. [26] also introduced instance-related representations to empower the model. Inspired by the progress of those multimodal learning works, we make the first attempt to leverage linguistic information to facilitate the vessel keypoint detection of CAG images.

### B. Vessel Segmentation

During the past decades, vessel segmentation has been attracting much attention in medical image analysis tasks as it

is crucial for developing intelligent diagnosis and treatment systems. Deep learning methods have been dominant since the emergence of U-Net [27], which has been adopted by many segmentation models as the backbone [28], [29], [30], [31]. Zhang et al. [30] designed an edge-gated flow to make U-Net more sensitive to vessel edges. Tan et al. [31] introduced prior knowledge of the vascular skeleton to improve the completeness and continuity of vessel segmentation results. However, due to the contradiction between the workload of complex vessel annotation and the model's dependence on large data, the generalization ability of supervised methods is not satisfactory.

Independent from the labeled segmentation masks, unsupervised methods are mainly based on manually designed features [32], [33], [34]. Memari et al. [33] proposed to enhance the image first with contrast limited adaptive histogram equalization (CLAHE) and Gabor and Frangi filters, then apply a genetic algorithm to obtain vessel masks. Kuppusamy et al. [34] applied Gabor and Canny edge detector and used random forest to ensemble the features, resulting in better detection of vascular structures. Based on the homogeneity of pixels, region growing is a common technique adopted in vessel segmentation tasks [35], [36], [37]. Zeng et al. [35] applied region growing to obtain thin vessel segmentation, combined the active contour model with a clustering algorithm to get thick vessel segmentation, and merged them to generate final results. Rodrigues et al. [36] proposed a framework consisting of two parts, which first recognized interested regions from 35 kinds of feature maps with machine learning classification algorithms, then took points with maximal connected response to be seeds and applied region growing to obtain vessels. Fu et al. [37] first designed a total variation regularized robust principal component analysis algorithm to extract vessels from the noisy background, then applied two-stage region growing to realize coronary vessel segmentation. Since unsupervised methods are based on low-level pixel features, they often lack the necessary semantic information, which is significant for extracting main vessel segments. Therefore, we proposed a supervision-reduced approach to combine the low-level grayscale feature with high-level semantic information (i.e., corresponding endpoints), to achieve accurate and flexible vascular anatomical segmentation.

## III. METHODOLOGY

The schematic of the proposed framework is shown in Fig. 2, which mainly consists of four steps. This section will present the details of each step in turn, including procedure interaction for acquiring and standardizing linguistic information (Section III-A), the LEED network for detecting endpoints of each vessel (Section III-B), the TS-Seg algorithm for obtaining segmentation anatomical masks (Section III-C), and vessel analysis for generating a complete vessel analysis report (Section III-D).

### A. Introduction of the Surgery Interaction

The communication among clinicians greatly reflects the progress of the surgery and contains crucial surgical information, thus leveraging this information can compensate for the limitations of image-only analysis. The illustration of the

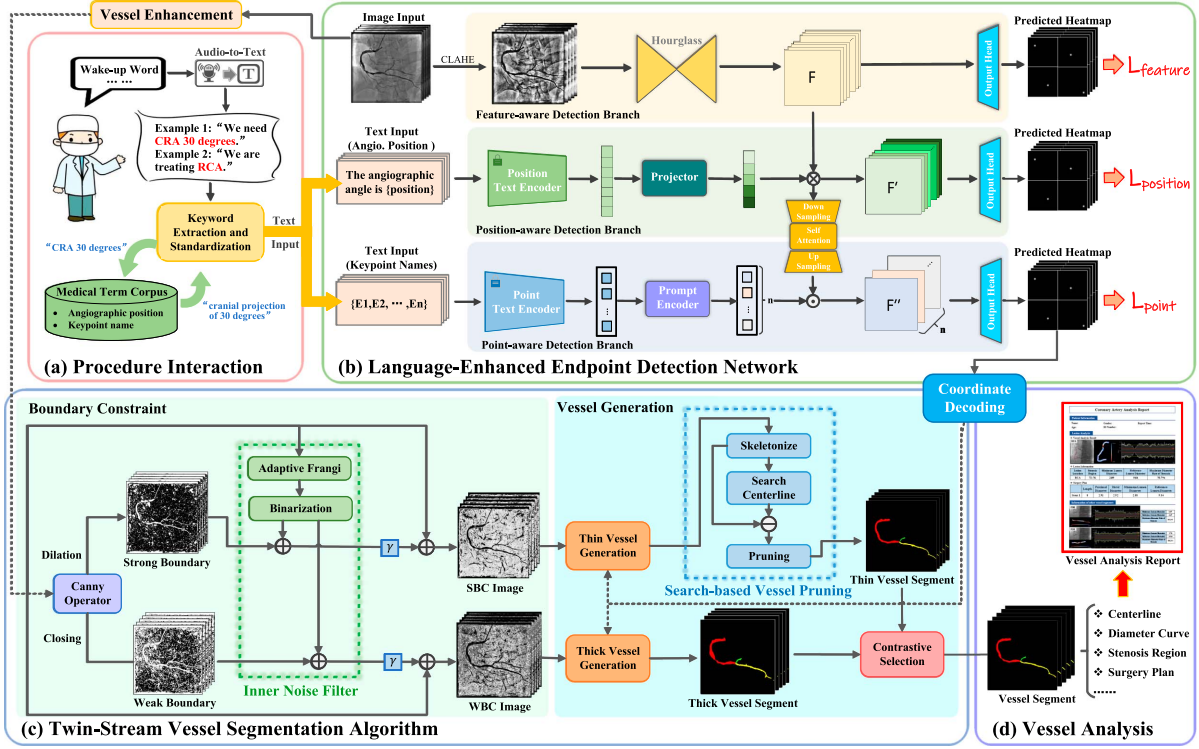


Fig. 2. Schematic of the proposed language-enhanced supervision-reduced vessel analysis framework. (a) Procedure interaction for acquiring and standardizing input conversation. (b) LEED for detecting endpoints of each vessel segment. (c) TS-Seg algorithm for obtaining segmentation anatomical results based on the endpoint positions provided by LEED. (d) Vessel analysis for generating complete vessel analysis reports based on the segmentation results obtained from TS-Seg.

surgery interaction module is shown in Fig. 2(a). Inspired by current AI assistants such as Siri, we have implemented a wake-up word for enabling interactive engagement with clinicians during operations. Once triggered, any subsequent speech is converted into text through an applicable speech-to-text tool.

Given that the two most crucial types of information during vascular interventional procedures are the angiographic position and the target vessel, a structured medical term corpus is constructed to include and convey that information. The contents, which are shown in Table I, are curated according to standard clinical practice and guidelines. Specifically, for angiographic positions, we chose six common angles for observing main coronary arteries [38], excluding those where some endpoints cannot be seen; for vessels, we distinguished between major coronary systems (left and right coronary arteries) for their substantial structural differences [39]. Instead of relying on unconstrained free language, the available inputs of the corpus are the keywords that clinicians will mention, typically some abbreviations, while the standardized outputs are the full names of the angiographic positions, as well as the endpoint names of the corresponding vascular branches. Considering that different clinicians may use varying expressions for the same terminology, multiple available input forms are assigned to each standardized output to ensure robust recognition. Once any one of the available input keywords is contained in the clinicians' words, it will be recognized and mapped to the corresponding standardized output text by the devised keyword extraction and

standardization module. Instead, any irrelevant conversation or acoustic noise will be ignored, causing no effect on the downstream network. Although Table I lists only two examples for each entry, the corpus is easily expandable to include additional synonyms, multilingual expressions, and region-specific habits without modifying the workflow. Importantly, because the outputs remain standardized regardless of input variation, it maintains consistent linguistic representations. Through this design, clinicians can interact naturally during the procedure by simply including a wake-up word in their routine conversation, while the model robustly interprets the linguistic cues for downstream algorithms.

### B. Overview of the LEED

Fig. 2(b) illustrates the structure of the proposed LEED network. The LEED network adopts a hierarchical architecture consisting of three components: feature-aware detection branch, position-aware detection branch, and point-aware detection branch. The feature-aware detection branch employs convolutional layers to extract visual features solely from the input CAG images. The other two branches are designed to progressively integrate linguistic information with visual representations. According to the clinical characteristics of the vascular interventional procedure, we utilize linguistic information from two perspectives: the angiographic position level and the point level. In the position-aware detection branch, the angiographic

TABLE I  
DETAILS OF THE CONSTRUCTED MEDICAL TERM CORPUS

| Information Type      | Available Input       | Standardized Output                                                                                                                                                                           |
|-----------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Angiographic Position | CRA 30°               | Cranial projection of 30°                                                                                                                                                                     |
|                       | CRA 30                |                                                                                                                                                                                               |
|                       | LAO 30°               | Left anterior oblique projection of 30°                                                                                                                                                       |
|                       | LAO 30                |                                                                                                                                                                                               |
|                       | CAU 20°               | Caudal projection of 20°                                                                                                                                                                      |
|                       | CAU 20                |                                                                                                                                                                                               |
|                       | LAO 45° and CRA 20°   | Left anterior oblique projection of 45° and cranial projection of 20°                                                                                                                         |
|                       | LAO 45 and CRA 20     |                                                                                                                                                                                               |
|                       | RAO 30° and CAU 20°   | Right anterior oblique projection of 30° and caudal projection of 20°                                                                                                                         |
|                       | RAO 30 and CAU 20     |                                                                                                                                                                                               |
|                       | RAO 30° and CRA 20°   | Right anterior oblique projection of 30° and cranial projection of 20°                                                                                                                        |
|                       | RAO 30 and CRA 20     |                                                                                                                                                                                               |
| Vessel Type           | Right coronary artery | {The proximal point of right coronary artery,<br>the distal point of right coronary artery,<br>the distal point of posterior descending artery,<br>the distal point of posterolateral branch} |
|                       | RCA                   |                                                                                                                                                                                               |
|                       | Left coronary artery  | {The bifurcation point,<br>the distal point of left circumflex coronary artery,<br>the distal point of left anterior descending artery}                                                       |
|                       | LCA                   |                                                                                                                                                                                               |

position provides a description of the entire image, providing global guidance related to a specific position. In the point-aware detection branch, the endpoint names provide regional information, allowing the model to achieve fine-grained spatial alignment with target keypoints.

In the feature-aware detection branch, for the original input CAG image  $I \in \mathbb{R}^{3 \times H \times W}$ , we first applied the bilateral filter and CLAHE to enhance vessel structure while suppressing background noise. The hourglass network [14] is applied to extract multiscale image features to generate the original feature map  $F \in \mathbb{R}^{C_e \times H \times W}$ , where  $C_e$  is the embedding dimension. As  $F$  is entirely obtained through convolution operations, it is necessary to integrate the captured surgical linguistic information to complement image features.

Since the descriptions are all converted from voice to text in natural languages, pretrained CLIP text encoders [22] are adopted in both position- and point-aware detection branches to exploit the related features in the textual modality. Note that we have frozen the parameters of both CLIP encoders to preserve the learned knowledge from pretraining. Given the esoteric nature of the description of angiographic positions, the pretrained CLIP encoder may struggle to effectively extract the semantic representations. Consequently, we opt to enrich the context information as much as possible by constructing the position-specific text prompt for each image  $I_i$  with the following template:

$$\text{Prompt}_i = \text{The angiographic angle is \{position}_i\} \quad (1)$$

where  $\text{position}_i$  is the standardized full terminology of the current angiographic position  $i$  as mentioned in Table I. The prompt first passes through a pretrained CLIP encoder to

get the matching representation  $R$ . Then, the position representation  $R$  will be mapped into the task-adaptive multimodal space with a projector. In this task, we adopted a lightweight structure to enable efficient task adaptation with minimal risk of distorting or overriding the pretrained representations, which is formulated as follows:

$$R_p = \text{Sigmoid}(\text{Linear}(\text{ReLU}(\text{Linear}(R)))) \quad (2)$$

where  $R_p \in \mathbb{R}^{1 \times C_e}$  is the projected representation. After projection,  $R_p$  is elementwise multiplied with the original feature map  $F$ , thereby amplifying the most relevant channels associated with the particular position. The formula is as follows:

$$F' = R_p^T \times F \in \mathbb{R}^{C_e \times H \times W}. \quad (3)$$

This process enhances the model's recognition ability of feature points under the specific pattern corresponding to the angiographic position.

After obtaining the feature map  $F'$  that has been merged with position information, knowledge at the point level is added to further refine the representation of endpoints. Specifically, the names of  $N$  endpoints will be turned into point embeddings by the CLIP pretrained encoder.  $N$  point embeddings corresponding to different endpoints are sent into the model simultaneously. To further enhance the spatial representations among different points, we employ a prompt encoder consisting of a single-layer transformer and a linear layer to model the long-range relationship and project the point embeddings into the multimodal feature space to get point feature vectors  $V_{\text{Point}} \in \mathbb{R}^{N \times C_e}$ . For the merged feature map  $F'$ , it is first compressed to a smaller dimension through downsampling, then adapted to point-level features via a self-attention module, and finally

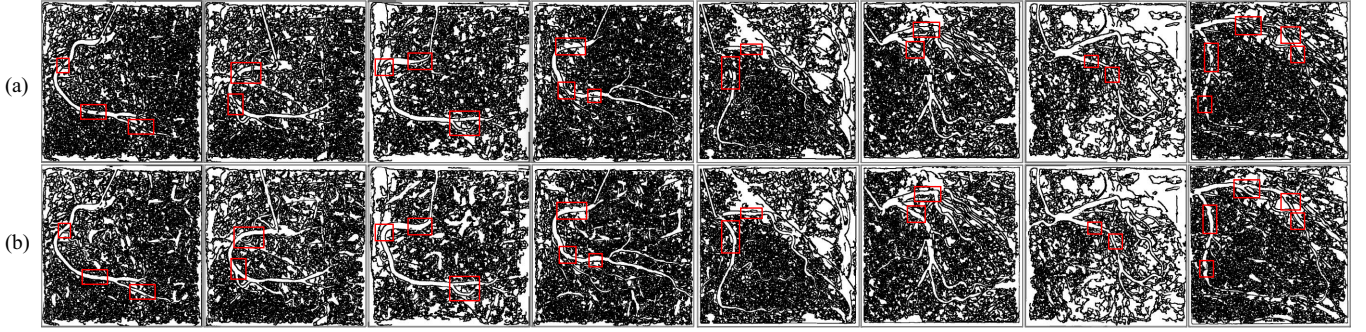


Fig. 3. Effects of the INF. (a) Boundary image before INF. (b) Boundary image after INF.

restored to its original dimension through upsampling. In this way, we can get the spatial probability map of each endpoint  $F''$  through inner product operation between point feature vectors and adapted feature maps  $F'_a \in \mathbb{R}^{C_e \times H \times W}$ . The formulations are as follows:

$$F'_a = \text{Up}(\text{Self\_Attn}(\text{Down}(F'))) \quad (4)$$

$$F''_i = V_{\text{Point}}^i \cdot F'_a, i = 1, 2, \dots, N \quad (5)$$

where  $F''_i \in \mathbb{R}^{H \times W}$  is the score map for endpoint  $i$ , and  $V_{\text{Point}}^i \in \mathbb{R}^{1 \times C_e}$ .

In each branch, an independent output head is attached to map channel numbers of feature maps ( $F, F', F''$ ) to key-point numbers for heatmap regression. The generated heatmaps from all three branches are individually supervised using MSE Loss. Specifically, the feature-, position-, and point-aware detection branches each predict a heatmap  $y_{i,j,n}^{(k)}$  where  $k \in \{\text{feature}, \text{position}, \text{point}\}$  denote different branches, and each is optimized against the same ground-truth heatmap  $\hat{y}_{i,j,n}$ . The loss for each branch is formulated as

$$\mathcal{L}_k = \sum_{i,j,n} (y_{i,j,n}^{(k)} - \hat{y}_{i,j,n})^2 \quad (6)$$

$i = 1, 2, \dots, H; j = 1, 2, \dots, W; n = 1, 2, \dots, N$

where  $y_{i,j,n}$  and  $\hat{y}_{i,j,n}^{(k)}$  denote the predicted and ground-truth heatmap values for the  $n$ th endpoint at spatial location  $(i, j)$ . Thus, the overall training loss can be written as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{point}} + \alpha_1 \cdot \mathcal{L}_{\text{position}} + \alpha_2 \cdot \mathcal{L}_{\text{feature}} \quad (7)$$

where  $\alpha_1$  and  $\alpha_2$  are two hyperparameters for adjusting the contribution of each branch. They are set to be 1.0 and 0.8 in our experiments. With such an ensemble, the medical knowledge obtained from different modalities can be fused to achieve a more informative representation for each endpoint.

### C. Details of the TS-Seg Algorithm

As shown in Fig. 2(c), the proposed TS-Seg algorithm mainly consists of two parts, including boundary constraint and vessel generation. Details of each part will be introduced in this section.

1) *Boundary Constraint*: The original CAG image is first denoised by a bilateral filter and enhanced by CLAHE, then the Canny operator [40] is adopted to obtain the original boundaries of the vessels. To bridge large gaps along the boundaries, morphological dilation operation is applied to generate a strong boundary map. Since the dilation operation can cause a thinner vascular region, we add another branch to utilize the morphological closing operation to retain information near boundaries, resulting in a weak boundary map.

Due to the uneven distribution of the contrast agent, some lines within the vessel lumen are detected as well, as shown in Fig. 3(a). Thus, an INF is designed to eliminate them. Frangi filter is a common method for heightening tubular structures [41]. However, vessel structures are best detected when the Frangi filter's scale matches the vessel diameter. Thus, directly applying the Frangi filter with fixed parameters on the whole image may lead to the omission of some thin vessel segments. To address this, we propose an adaptive Frangi: dividing the whole image into several patches ( $2 \times 2$  in this article) and applying Frangi filter within each one. After that, Otsu thresholding method [42] is employed to achieve binarization, resulting in obtaining a relatively complete vascular region. By adding it to the strong and weak boundary maps, the inner noise can be effectively eliminated without destroying any boundaries, as shown in Fig. 3(b).

At last, after being multiplied by a weight factor  $\gamma$  and normalized to the range of 0–255, the filtered strong and weak boundary maps are added to the original enhanced image to obtain strong boundary-constrained (SBC) and weak boundary-constrained (WBC) images, respectively, as presented in Fig. 4.  $\gamma$  is set to be 3 in this experiment. Note that both SBC and WBC images will be inverted to make the images conform to the common forms, that is, the blood vessel pixel being 255 while the background pixel being 0.

2) *Vessel Generation*: After obtaining the boundary-constrained images, we adopted the coordinate decoding algorithm [43] to decode the endpoint coordinates from the heatmaps predicted by LEED. Based on the starting and ending points of each vessel branch, region growing is adopted on the SBC and WBC images separately. The thin vessel segment obtained from the SBC image can substantially remove superfluous branches and reduce over-growth caused by discontinuous boundaries but may appear narrower than ground truth due

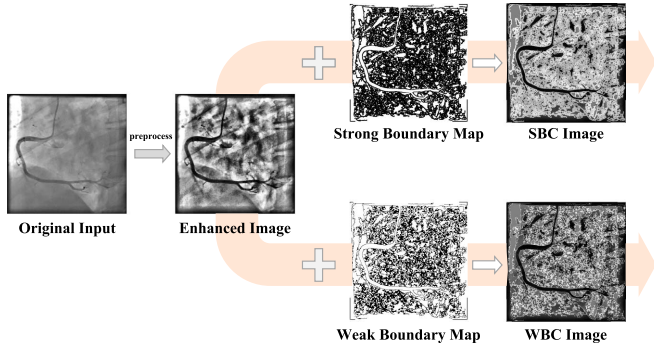


Fig. 4. Difference between two boundary-constrained images.

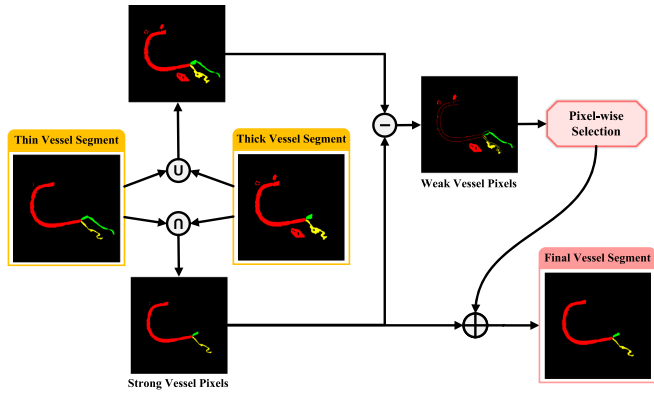


Fig. 5. Diagram of the contrastive selection algorithm.

to boundary region loss. Conversely, the thick vessel segment grown from the WBC image can more accurately restore the boundary regions, though it may include unnecessary vessel branches or experience significant over-growth.

To integrate the advantages of both images, we propose a contrastive selection algorithm as illustrated in Fig. 5. The core idea is that foreground pixels appearing in both thin and thick vessel segment images are more likely to be true vessel pixels, whereas those appearing in only one image require further verification. Therefore, the intersection of the two images is taken as strong vessel pixels, while the union of them minus the intersection is considered as weak vessel pixels. For each weak vessel pixel, if the difference between its grayscale and the mean value of the strong vessel pixels is less than a threshold  $\beta$ , it will be considered as a selected vessel pixel; otherwise, it will be classified as background. The final vessel segmentation results are formed by the combination of strong vessel pixels and selected vessel pixels. The grayscales are obtained from the original enhanced image. In this article,  $\beta$  is set to 5 for the RCAE dataset and 20 for the LCAE dataset.

Noticing that some thick vessel branches with a similar grayscale to the main vessels may not be eliminated by the previous steps, we devise a SVP algorithm as shown in Fig. 6. For each vessel branch, the first step is to ensure the two endpoints are in the same connected domain. If not, the closest point pair  $(P'_s, P'_e)$  of the two connected domains where each endpoint

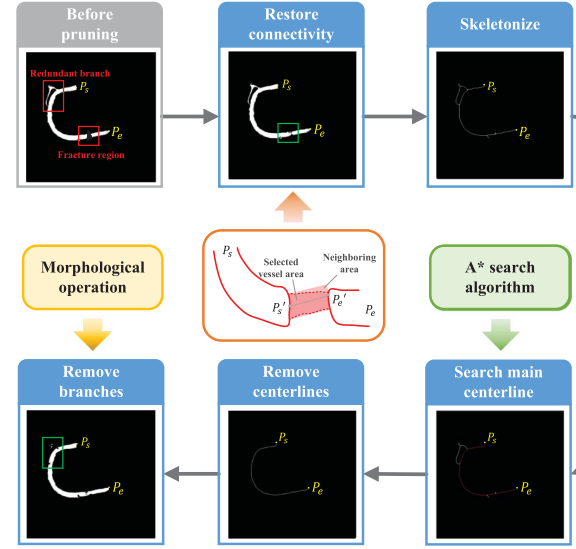


Fig. 6. Diagram of the SVP algorithm.

resides will be collected. Then, for each pixel  $p$  in the neighboring area around the line  $P'_s P'_e$ , the minimum absolute difference with the two endpoints  $MAD(p, P_s, P_e)$  will be calculated as follows:

$$MAD(p, P_s, P_e) = \min\{|G(p) - G(P_s)|, |G(p) - G(P_e)|\} \quad (8)$$

where  $G(\cdot)$  refers to the grayscale of the pixel. If  $MAD(p, P_s, P_e)$  is less than the threshold  $\theta$ , pixel  $p$  will be classified as the vessel. To make the process more adaptive, if the endpoints remain disconnected after one iteration,  $\theta$  will be added by 5, and the aforementioned process will repeat until the endpoints are in the same connected domain.  $\theta$  is set to be 10 initially in this article. After restoring the vessel connectivity, the vessel segment will be skeletonized to obtain the centerlines. Then, the A\* algorithm [44] is utilized to search the shortest path between the two endpoints along the centerline, thus obtaining the centerline of the main vessel. Considering that the endpoints may not be precisely located within the centerline, the starting and ending points for the search algorithm are set to the centerline points that are closest to the endpoints. By subtracting the main centerline from the original one, the redundant branches can be identified and removed using morphological operations to achieve pruning. The pruning results can be seen in Fig. 7.

#### D. Vessel Analysis and Report Generation

To directly enable the proposed framework to be helpful in clinical scenarios, comprehensive vessel analysis reports are provided based on the anatomical segmentation results. Examples of coronary artery analysis reports before and after the surgery are shown in Fig. 8, where the CAG images are acquired from a real patient. For each vascular branch, the central and contour lines are first extracted from the anatomical segmentation results, followed by the computation of the

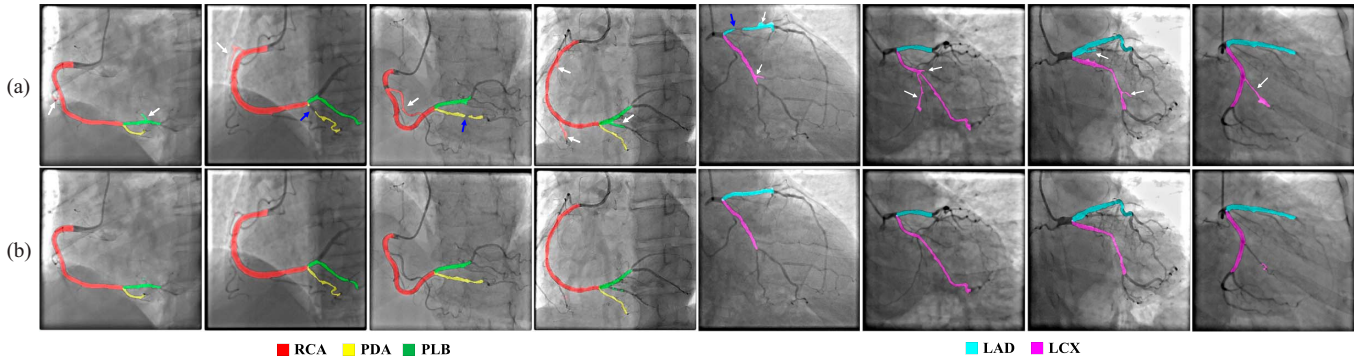


Fig. 7. Effects of the search-based vessel pruning. (a) Thin vessel segments before pruning. (b) Thin vessel segments after pruning. White arrows point out the pruned unnecessary branches. Blue arrows point out the fractured areas that have been connected.

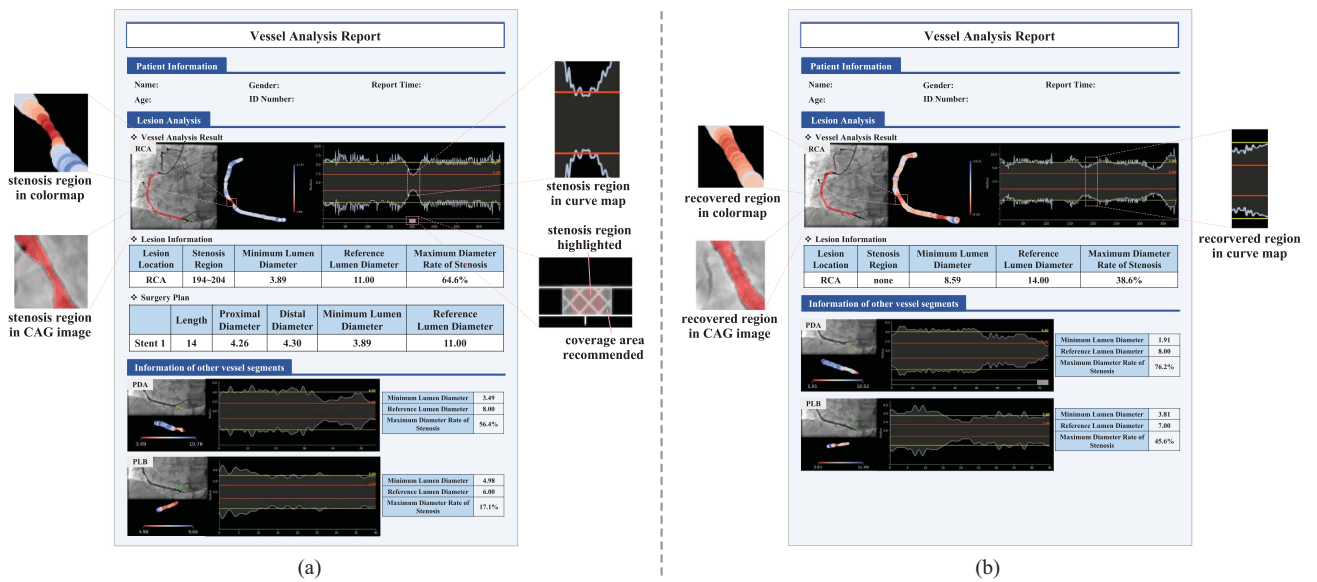


Fig. 8. Coronary artery analysis reports of a case. (a) Preoperative report. (b) Postoperative report.

normal direction at each point in the centerline. In this way, the vessel lumen diameter at each point can be easily acquired, allowing the calculation of the reference diameter and lesion information. Specifically, considering the mean value of the diameters could be skewed by anomalously small values at the stenosis area, the median across all points is chosen as the reference diameter. According to [45], coronary arteries with significant stenosis ( $\geq 70\%$  diameter) require surgical intervention. To prevent overlooking potential stenosis, we select the 60% stenosis rate as the threshold for determining the presence of significant ones. An area with a diameter smaller than the stenosis threshold is considered to be a stenosis lesion.

A standard report consists of three parts, including patient information, lesion analysis, and information on other vessel segments. The patient information module is for obtaining basic personal information of the patients, following the conventional medical examination report format. The lesion analysis module presents the details of the vessel segment with detected lesions. First, the original CAG image with enhanced target vessel and

its 3D diameter colormap are displayed, where red indicates a smaller diameter and blue means larger ones, allowing clinicians to directly identify the location and morphology of stenosis in a visual manner. To further support clinical decision-making, we visualize the diameter curve map and highlight the reference lumen diameter and stenosis threshold by yellow and red lines separately in the graph. Note that the numbers on the Y-axis in the graph are all radius values. Moreover, for diseased vessels, the stenosis region and the area recommended for stent coverage are highlighted beneath the diameter curve. According to the calculated lesion information, such as stenosis length and minimum lumen diameter, the report provides suggested key quantitative values required for selecting the appropriate stent, such as length and proximal/distal diameters. All the contents in the report are exactly the ones that clinicians need to measure manually in current practice. Thus, the generated report shows great potential in reducing measurement workload and improving decision efficiency. Information on other vessel segments that do not currently require intervention is also provided, such

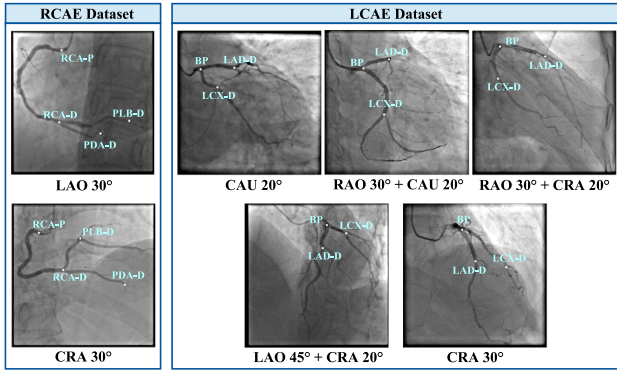


Fig. 9. Angiographic position information of the datasets.

as the maximum stenosis rates, as a reference for physicians and patients to apply monitoring or preventative care.

By comparing preoperative and postoperative reports, the therapeutic effectiveness of the surgery can also be evaluated. Comparing Fig. 8(a) and (b), there exists a significant stenosis area in the patient's RCA before the surgery in Fig. 8(a), while the vessel diameters in the corresponding area are restored to the reference diameter after the intervention in Fig. 8(b), indicating a successful therapeutic outcome. It should be noted that the “stenosis” marked in PDA in Fig. 8(b) actually represents a rapid diminution in diameters toward the end instead of pathological abnormality.

#### IV. EXPERIMENTS AND RESULTS

##### A. Datasets

*RCAE dataset* is an in-house right coronary artery (RCA) endpoint dataset consisting of 2276 CAG images from 269 clinical cases. Four endpoints of three main vascular segments are marked, including the proximal and distal points of the RCA (namely RCA-P and RCA-D), the distal point of the posterior descending artery (PDA-D), and the posterolateral branch (PLB-D). All the images are labeled with the corresponding angiographic positions, which are either  $CRA\ 30^\circ$  or  $LAO\ 30^\circ$ . To ensure the validity of the experimental evaluation, the data of the training set (1462 images), validation set (414 images), and testing set (400 images) are independent of each other. All the images are reshaped into  $224 \times 224$  pixels.

*LCAE dataset* is an in-house left coronary artery endpoint dataset containing 3225 CAG images from 219 clinical cases. Three endpoints of two main vessels are marked, including the bifurcation point (BP) of the LAD artery and LCX coronary artery, and the distal points of them (namely LAD-D and LCX-D). All the images are labeled with the corresponding angiographic positions, which include five categories as shown in Fig. 9. The data of the training set (2071 images), validation set (585 images), and testing set (569 images) are independent of each other. All the images are reshaped into  $224 \times 224$  pixels.

*XCAD dataset* is a publicly available dataset that contains 126 CAG images along with segmentation masks [46]. We divided the dataset into two subsets: XCAD-L for the left

coronary artery and XCAD-R for the RCA, in line with our experiment settings. After excluding the images that correspond to angiographic positions that failed to present the target vascular branches we needed, there were 51 images for the XCAD-L and 23 images for the XCAD-R. After that, we asked clinicians to annotate the keypoints and angiographic positions of the images in this dataset. Given the limited size of the public dataset, we decided to follow [12] to train the deep learning models on our dataset and test them on the XCAD. For multibranching vessel segmentation tasks, we removed the collateral vessel label from the original segmentation mask and retained the target vascular branches, i.e., RCA, PDA, and PLB for the RCA, and LAD and LCX for the left coronary artery.

##### B. Implementation Details and Evaluation Metrics

The LEED is implemented on the PyTorch library (version 1.13.1) with one NVIDIA Tesla A6000 GPU. The batch size is 16, and the maximum epoch number is 100 during the training stage. AdamW is used as the optimizer with an initial learning rate of 0.001, which decays exponentially with gamma of 0.9.

For the assessment of LEED, drawing upon studies in key-point detection, we employ the percentage correct keypoints (PCKs) as the evaluation metric in this task and set the threshold to be 15px. The Euclidean distance between the predicted point and the ground truth, i.e., the average pixel distance (APD), is also calculated as a reference. For the evaluation of the TS-Seg, common metrics for segmentation tasks such as F1-score and intersection of union (IoU) are adopted.

Due to the fact that there exist all kinds of complicated interference in real CAG images [3], to objectively evaluate its standalone performance, we adopt the ground-truth endpoint coordinates as the input to the TS-Seg algorithm in all experiments, unless otherwise specified. The real-world usability of the TS-Seg algorithm will be independently evaluated with the predicted endpoint coordinates as input.

##### C. Results of the LEED Network

*1) Comparison With Other Methods:* To demonstrate the advantage of our LEED network, we compare it with several state-of-the-art keypoint detection methods, including MSPN [47], SG-Posenet [48], HRNetV2-W18 [17], HigherHRNet-W32 [18], HRFormer-S [19], SimCC [49], FARNet [50], and MNet [51]. Since the compared models are originally designed to generate heatmaps with different resolutions, we added a single upsampling layer at the end of the models with a smaller output to unify the output size across different methods. Except for this, all the models remain the original structure and settings in their official open-source implementations. Table II clearly shows that LEED achieves the best overall performance in terms of mean PCK and APD on both RCAE and LCAE datasets, with an advantage in parameters. Particularly, our approach is far ahead in almost every evaluation metric on the LCAE dataset, achieving an improvement of PCK by 3.1% and APD by 1.61 over the second-best HRFormer-S. P-values from the paired Wilcoxon test between the LEED and other methods are provided in Table II. As can be seen, most p-values are

TABLE II  
COMPARISON WITH STATE-OF-THE-ART KEYPOINT DETECTION NETWORKS

| Methods         | RCAE Dataset |             |              |              |              |              | LCAE Dataset |              |              |             |              | Params.<br>(M) | FLOPs<br>(G) |
|-----------------|--------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|--------------|----------------|--------------|
|                 | APD↓         |             |              |              |              | PCK↑         | APD↓         |              |              |             | PCK↑         |                |              |
|                 | RCA-P        | RCA-D       | PDA-D        | PLB-D        | Mean         |              | BP           | LCX-D        | LAD-D        | Mean        |              |                |              |
| MSPN            | 5.96         | 4.28        | <b>25.43</b> | 14.99        | 13.34        | 0.740        | 2.94         | 16.55        | 14.04        | 11.18       | 0.735        | 56.63          | 11.39        |
|                 | (5.58e−02)   | (8.30e−06)  | (5.88e−04)   | (3.49e−01)   |              | (2.16e−04)   | (8.14e−26)   | (3.43e−14)   | (1.33e−14)   |             | (5.05e−09)   |                |              |
| SG-Posenet      | <u>5.47</u>  | 3.64        | 28.47        | <b>14.31</b> | <u>12.98</u> | 0.736        | 2.35         | 16.51        | 17.08        | 11.98       | 0.766        | 10.61          | 169.59       |
|                 | (2.07e−02)   | (3.09e−04)  | (2.61e−01)   | (1.09e−01)   |              | (6.56e−04)   | (8.16e−07)   | (3.08e−05)   | (4.78e−12)   |             | (4.28e−06)   |                |              |
| HRNetV2-W18     | 6.67         | <u>2.99</u> | 29.20        | 16.77        | 13.90        | <u>0.753</u> | 1.26         | 16.20        | 15.18        | 10.88       | 0.747        | 3.54           | 2.21         |
|                 | (5.12e−01)   | (3.63e−03)  | (2.26e−03)   | (2.98e−04)   |              | (6.86e−03)   | (8.68e−04)   | (2.24e−10)   | (1.06e−10)   |             | (4.01e−07)   |                |              |
| HigherHRNet-W32 | 7.89         | 4.61        | 34.99        | 18.64        | 16.53        | 0.689        | 2.81         | 22.82        | 21.41        | 15.68       | 0.710        | 10.25          | 9.15         |
|                 | (4.31e−02)   | (4.67e−09)  | (5.89e−05)   | (6.74e−06)   |              | (8.89e−09)   | (4.83e−10)   | (1.74e−15)   | (1.32e−12)   |             | (3.95e−12)   |                |              |
| HRFormer-S      | <b>5.36</b>  | 4.06        | <u>27.25</u> | 18.95        | 13.9         | 0.751        | 2.55         | <u>12.57</u> | 13.06        | <u>9.39</u> | <u>0.787</u> | 6.00           | 2.15         |
|                 | (8.20e−01)   | (1.50e−09)  | (1.33e−02)   | (7.72e−04)   |              | (9.52e−03)   | (1.27e−03)   | (2.88e−03)   | (3.82e−09)   |             | (1.27e−03)   |                |              |
| SimCC           | 11.41        | 9.40        | 32.49        | 33.76        | 21.76        | 0.521        | 4.95         | 26.76        | 18.81        | 16.84       | 0.644        | 25.70          | 2.03         |
|                 | (3.25e−12)   | (4.21e−16)  | (7.53e−09)   | (5.76e−13)   |              | (7.23e−23)   | (5.00e−86)   | (1.94e−41)   | (2.21e−37)   |             | (9.39e−35)   |                |              |
| FARNet          | 6.68         | 7.10        | 25.99        | 20.60        | 15.09        | 0.714        | 1.60         | 15.94        | <u>12.32</u> | 9.96        | 0.756        | 79.02          | 13.79        |
|                 | (9.59e−01)   | (1.97e−13)  | (1.61e−04)   | (3.16e−08)   |              | (1.18e−06)   | (5.67e−18)   | (3.79e−11)   | (3.11e−06)   |             | (1.75e−07)   |                |              |
| MNet            | 5.87         | 6.30        | 28.71        | 23.42        | 16.07        | 0.702        | <b>1.24</b>  | 18.16        | 14.80        | 11.40       | 0.754        | 10.04          | 15.95        |
|                 | (2.92e−02)   | (5.85e−10)  | (1.19e−03)   | (2.94e−04)   |              | (6.50e−06)   | (5.08e−02)   | (5.90e−10)   | (4.43e−07)   |             | (1.05e−09)   |                |              |
| LEED (Ours)     | 7.13         | <b>2.20</b> | 27.38        | <u>14.71</u> | <b>12.85</b> | <b>0.770</b> | <u>1.25</u>  | <b>12.19</b> | <b>9.89</b>  | <b>7.78</b> | <b>0.818</b> | 8.59           | 59.37        |

Note: The bold font highlights the optimal results, while the underlined font indicates suboptimal ones. P-values are shown in brackets.

small ( $< 0.05$ ), indicating that the proposed LEED network is statistically significantly better than other comparative methods. On the more challenging RCAE dataset, which contains more endpoints and fewer distinct vascular branches, LEED still achieves notable gains of 1.7% and 0.13 in average PCK and APD, compared with suboptimal results. Yet, it does not outperform all baselines on certain points. This is because the LEED focuses on leveraging language information to improve global structural consistency across multiple endpoints, which benefits overall accuracy but may reduce flexibility in adapting to localized variations.

Furthermore, we conduct comparison experiments on the public XCAD dataset. The results are presented in Table III. As shown, due to the lack of fine-tuning and the lower image quality of the XCAD dataset, the performance of all methods is reduced compared to the results on our own dataset. However, our method achieves the second-best results for both the left and right coronary arteries. This could demonstrate the generalizability and robustness of the proposed LEED network across datasets with different characteristics.

2) *Ablation Study*: To comprehend the contribution of the language priors in the proposed LEED, we perform detailed ablation experiments by gradually adding each language-enhanced branch to the vision-based baseline. The results are presented in Table IV. When removing all linguistic inputs and relying solely on visual features, the model exhibits noticeably worse endpoint detection performance in terms of both PCK and APD. Introducing either positional information or endpoint-related descriptions individually yields clear performance gains, indicating that linguistic priors provide complementary cues that cannot be inferred from images alone. More importantly, combining both types of language information leads to a substantial improvement across all evaluation metrics compared with the visual-only baseline. Specifically, the full model achieves gains of 3.7% in PCK and 0.74 in APD on the RCAE dataset, and 3.9% in PCK and 1.81 in APD on the LCAE dataset. Moreover, model parameters and FLOPs

TABLE III  
COMPARISON EXPERIMENT WITH DIFFERENT METHODS ON THE XCAD DATASET

| Method          | XCAD-R       |              | XCAD-L       |              |
|-----------------|--------------|--------------|--------------|--------------|
|                 | PCK↑         | APD↓         | PCK↑         | APD↓         |
| MSPN            | 0.717        | 15.79        | 0.569        | 18.71        |
| SG-Posenet      | <b>0.772</b> | <u>13.91</u> | <b>0.673</b> | <b>17.15</b> |
| HRNetV2-W18     | 0.730        | 14.06        | 0.608        | 17.45        |
| HigherHRNet-W32 | 0.707        | <b>13.80</b> | 0.588        | 19.68        |
| HRFormer-S      | 0.674        | 15.67        | 0.556        | 21.43        |
| SimCC           | 0.534        | 19.01        | 0.431        | 27.45        |
| FARNet          | 0.707        | 15.82        | 0.549        | 23.38        |
| MNet            | 0.734        | 15.70        | 0.543        | 25.44        |
| LEED(Ours)      | <u>0.737</u> | 14.33        | <u>0.614</u> | <u>17.71</u> |

Note: The bold font highlights the optimal results, while the underlined font indicates suboptimal ones.

TABLE IV  
ABLATION STUDY: EXPERIMENT RESULTS ON ADDING LANGUAGE-ENHANCED BRANCHES TO THE BASELINE ON TWO DATASETS

| Method               | RCAE Dataset |              | LCAE Dataset |             | Params.<br>(M) | FLOPs<br>(G) |
|----------------------|--------------|--------------|--------------|-------------|----------------|--------------|
|                      | PCK↑         | APD↓         | PCK↑         | APD↓        |                |              |
| Baseline             | 0.733        | 13.59        | 0.779        | 9.59        | 3.24           | 42.86        |
| + Pos                | 0.761        | 13.05        | 0.783        | 9.58        | 3.95           | 47.29        |
| + Point              | 0.744        | 13.19        | 0.790        | 9.04        | 8.53           | 54.94        |
| + Pos + Point (LEED) | <b>0.770</b> | <b>12.85</b> | <b>0.818</b> | <b>7.78</b> | 8.59           | 59.37        |

Note: "Baseline" means only using feature-aware detection branch, which is a vision-based model without linguistic guidance. "Pos" refers to position-aware detection branch, and "Point" refers to point-aware detection branch. Bold entries indicate the optimal results.

are calculated for each ablation variant. As summarized in Table IV, incorporating the two linguistic branches introduces only a marginal increase in parameter count and computational cost, while yielding substantial improvements in keypoint detection performance. These results clearly validate the essential

TABLE V  
ABLATION STUDY: EXPERIMENT RESULTS OF DIFFERENT TEXT PROMPTS  
FOR ANGIOGRAPHIC POSITIONS ON TWO DATASETS

| Prompt Setting                      | RCAE Dataset |              | LCAE Dataset |             |
|-------------------------------------|--------------|--------------|--------------|-------------|
|                                     | PCK↑         | APD↓         | PCK↑         | APD↓        |
| One-hot                             | 0.734        | 13.95        | 0.789        | 8.51        |
| {position}                          | 0.755        | 13.46        | 0.599        | 18.87       |
| Template + {position abbrev.}       | 0.756        | 13.37        | 0.603        | 18.52       |
| <b>Template + {position} (Ours)</b> | <b>0.770</b> | <b>12.85</b> | <b>0.818</b> | <b>7.78</b> |

Note: “Template” refers to the specific phrase “The angiographic angle is.”  
Bold entries indicate the optimal results.

role of the two linguistic branches in strengthening visual understanding.

Another ablation study is conducted to prove that the text prompt setting in (1) for angiographic positions is necessary. Results in Table V reveal that using one-hot encoding method nearly makes no difference compared with the baseline experiment in Table IV, which is in expectation since it does not contain any semantic information. Compared to the other three prompt settings containing position information, our setting performs the best, proving that providing comprehensive explanations of the prior position information is helpful for pretrained CLIP to extract effective knowledge representations. In this way, we can enhance the model’s understanding of the information conveyed in clinicians’ languages by designing appropriate text prompt settings, without requiring any modifications to the language inputs themselves. Clinicians do not need to change their clinical language habits for model performance. Yet, we observe that the other two settings could bring some improvement on the RCAE dataset, while showing poor performance on the LCAE dataset. We argue that this is attributable to the greater complexity of the positional information in the LCAE dataset, which encompasses more angles, especially including several combined angles, e.g., *LAO 45°* and *CRA 20°*. In such scenarios, due to its deficiency in domain-specific knowledge, the CLIP encoder may only be able to extract ambiguous or even confusing features without exhaustive and clear descriptions, leading to a degradation in model performance.

#### D. Results of the TS-Seg Algorithm

1) *Internal Validation*: Since existing unsupervised algorithms are only capable of segmenting all the vascular branches instead of specific ones, we choose to conduct the comparison experiment against supervised deep-learning-based methods. Specifically, we compare the proposed TS-Seg algorithm against five networks, including the classic UNet [27], advanced TransUNet [52], UCTransNet [53], DconnNet [54], and FTUNet [55]. All the methods are compared in their default settings without any modifications. The results are shown in Table VI. When using ground-truth endpoints as input, our method achieves superior performance compared with all competing approaches in terms of mean F1-score and mean IoU. When replacing the ground-truth endpoints with the ones predicted by LEED, although the performance shows a moderate decrease,

TABLE VI  
COMPARISON WITH SUPERVISED DEEP-LEARNING METHODS ON  
THE IN-HOUSE DATASETS

| Method     | RCAE Dataset |              | LCAE Dataset |              |
|------------|--------------|--------------|--------------|--------------|
|            | mF1          | mIoU         | mF1          | mIoU         |
| UNet       | 0.765        | 0.655        | 0.691        | 0.586        |
| TransUNet  | <u>0.813</u> | <u>0.716</u> | 0.749        | 0.656        |
| UCTransNet | 0.752        | 0.638        | 0.686        | 0.572        |
| DconnNet   | 0.807        | 0.706        | 0.705        | 0.602        |
| FTUNet     | 0.704        | 0.580        | 0.622        | 0.499        |
| Ours       | <b>0.845</b> | <b>0.736</b> | <b>0.879</b> | <b>0.785</b> |
| Ours*      | <u>0.813</u> | 0.709        | <u>0.772</u> | <u>0.687</u> |

Note: The bold font highlights the optimal results, while the underlined font indicates suboptimal ones. “Ours” and “Ours\*” refer to the results of the TS-Seg algorithm taking ground truth and predicted endpoint coordinates as the input, separately.

as expected due to endpoint localization error, the overall segmentation result still outperforms most baseline methods. This demonstrates that our approach remains robust under realistic conditions.

In addition, we further provide a qualitative visualization in Fig. 10. As illustrated, due to significant variations in vascular morphology, the segmentation results produced by the best two deep-learning methods suffer from fragmented vessel segments and interrupted branches. In contrast, our method generates vessel segments with more complete anatomical structures and fewer false-positive fragments. This advantage stems from the incorporation of boundary and keypoint constraints along with the SVP algorithm, which effectively removes redundant branches while preserving the region of the target branch. This provides a reliable foundation for downstream quantitative vessel analysis.

To directly evaluate the effectiveness of the two optimization algorithms, INF and SVP, an ablation study is conducted as shown in Table VII. It shows that the removal of either module leads to a deterioration in the outcomes across the two datasets, and removing both causes a serious performance drop in mean F1-score and mean IoU (0.120 and 0.140 for the RCAE dataset and 0.144 and 0.160 for the LCAE dataset), underlining the crucial role of INF and SVP. The reason why deleting INF could deteriorate the model performance more than removing SVP is that SVP is an optimization module for pruning obtained complete vessel segmentation results, while INF is a necessary preparation module, without which the growing vessel may be disconnected due to noisy lines within the vessel lumen, thus receiving poor results.

The final segmentation results along with the intermediate thin and thick vessel segmentation images are all presented in Fig. 11. It clearly illustrates that the thin vessel results more effectively eliminate the branching vessels compared to the thick vessel results, yet the vessels obtained are thinner than the actual vascular lumen. After applying our contrastive selection algorithm, the advantages of both types of images can be integrated to achieve superior performance. It proves

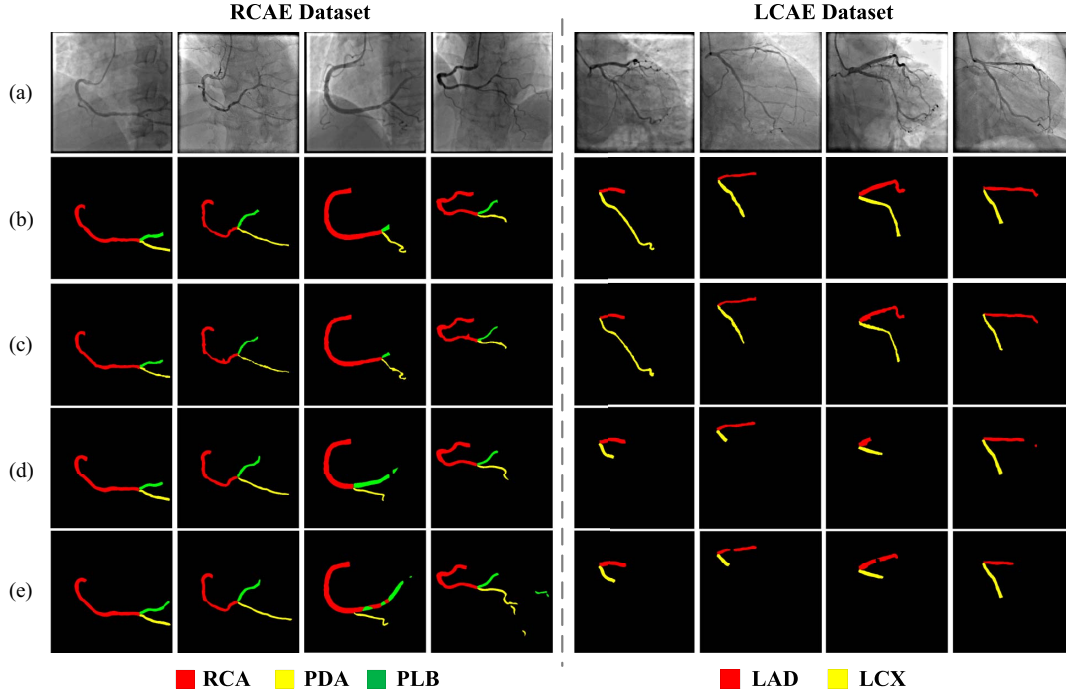


Fig. 10. Qualitative comparison results by different segmentation methods on the two in-house datasets. From top to bottom are: (a) Original CAG image; (b) Ground truth; (c) TS-Seg algorithm (ours); (d) TransUNet; and (e) DconnNet.

TABLE VII  
ABLATION STUDY FOR THE TS-SEG ALGORITHM

| Method             | RCAE Dataset |              |              |              |              |              |              |              | LCAE Dataset |              |              |              |              |              |
|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                    | RCA          |              | PDA          |              | PLB          |              | Mean         |              | LCX          |              | LAD          |              | Mean         |              |
|                    | F1           | IoU          | F1           | IoU          | F1           | IoU          | F1           | IoU          | F1           | IoU          | F1           | IoU          | F1           | IoU          |
| <b>Ours</b>        | <b>0.906</b> | <b>0.829</b> | <b>0.810</b> | <b>0.683</b> | <b>0.820</b> | <b>0.697</b> | <b>0.845</b> | <b>0.736</b> | <b>0.879</b> | <b>0.786</b> | <b>0.878</b> | <b>0.783</b> | <b>0.879</b> | <b>0.785</b> |
| Ours w/o INF       | 0.754        | 0.662        | 0.767        | 0.633        | 0.753        | 0.619        | 0.758        | 0.638        | 0.776        | 0.668        | 0.799        | 0.698        | 0.788        | 0.683        |
| Ours w/o SVP       | 0.861        | 0.761        | 0.778        | 0.641        | 0.801        | 0.671        | 0.813        | 0.691        | 0.845        | 0.736        | 0.866        | 0.766        | 0.856        | 0.751        |
| Ours w/o INF & SVP | 0.722        | 0.613        | 0.728        | 0.585        | 0.724        | 0.589        | 0.725        | 0.596        | 0.725        | 0.610        | 0.744        | 0.640        | 0.735        | 0.625        |

Note: Bold entries indicate the optimal results.

that our developed TS-Seg algorithm can successfully extract vessel segments simply based on endpoints on the two in-house coronary artery datasets.

2) *External Validation*: Additionally, we compare the segmentation performance on both in-house and public datasets, which is shown in Fig. 12. As can be seen, our method achieves an average F1 score of 0.802 on the XCAD-L and 0.821 on the XCAD-R, both of which are slightly lower than those on our own dataset. In terms of sensitivity, it shows a decrease of 0.026 compared to the performance on the LCAE dataset, while it demonstrates an increase of 0.03 regarding the RCA. Overall, the performance across all datasets exhibits no significant differences. This suggests that while there may be some variation due to the different characteristics of the datasets, the TS-Seg algorithm maintains a consistently strong performance across different datasets.

3) *Sensitivity Analysis*: To assess the robustness of TS-Seg with respect to hyperparameter variations, we conducted a comprehensive sensitivity analysis to analyze: the weight factor

$\gamma$ , the threshold  $\beta$  in the contrastive selection algorithm, and the threshold  $\theta$  in the SVP algorithm.

First, as shown in Fig. 13(a), once  $\gamma > 1$ , the model performance stabilizes and no longer changes with  $\gamma$ , demonstrating that the algorithm is insensitive to this parameter. Therefore, we fix  $\gamma = 3$  in all experiments. Second, as shown in Fig. 13(b), although varying  $\beta$  leads to slight fluctuations in performance, the variations consistently remain within 0.06 in terms of mF1, confirming that TS-Seg maintains strong robustness to this parameter. Third, as shown in Fig. 13(c), the initial value of  $\theta$  has no impact on the final segmentation results. This is because the connected-domain construction is an incremental and iterative process, where the initial value does not influence the final outcome. Instead, it affects the computation time per image: a smaller initial value results in more iterations, thereby increasing the overall processing time, as illustrated in Fig. 13(d). Thus, we set  $\theta$  to 10 for better computational efficiency. Overall, these experiments demonstrate that TS-Seg is robust to a wide range of parameter settings.

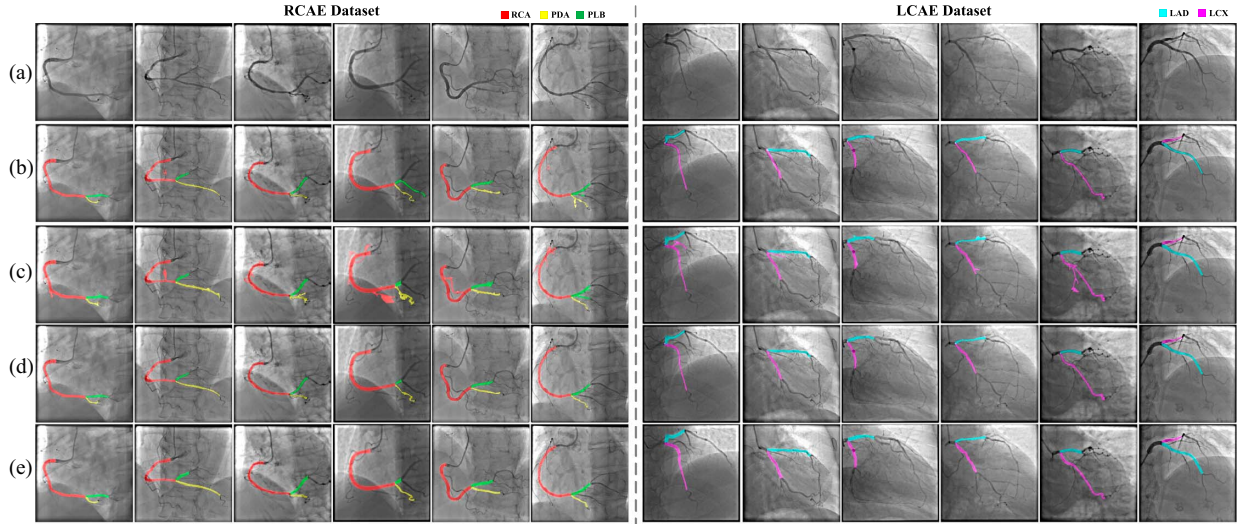


Fig. 11. Visualization results of the TS-Seg algorithm. Different colors represent different vessel segments as the legend shows. (a) Original CAG image. (b) Thin vessel segmentation result. (c) Thick vessel segmentation result. (d) Final vessel segmentation result. (e) Ground truth.

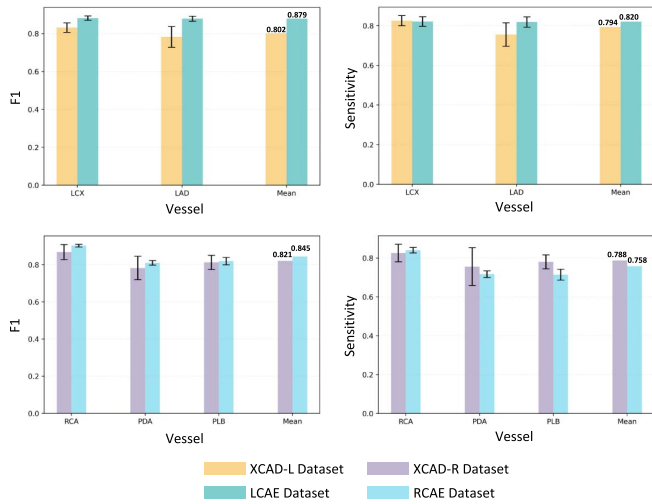


Fig. 12. Comparison results of the TS-Seg algorithm on different datasets.

4) *Flexibility Validation*: Considering that in real clinical surgeries, chances are that some patients have variant vessel branches that also contribute a lot to the blood supply system. It is difficult to apply deep-learning-based methods to analyze these vessel branches because the situation can be different for each individual, i.e., some have variant branches while others do not, making it hard to uniformly annotate. The proposed TS-Seg algorithm could solve the problem by simply manually selecting the endpoints of the variant vessel segment. Some examples are shown in Fig. 14, which clearly shows that with manually selected endpoints and the TS-Seg algorithm, those undefined variant vascular branches can also be accurately extracted. This makes our framework promising to adapt to various vessel structures in real-world cases.

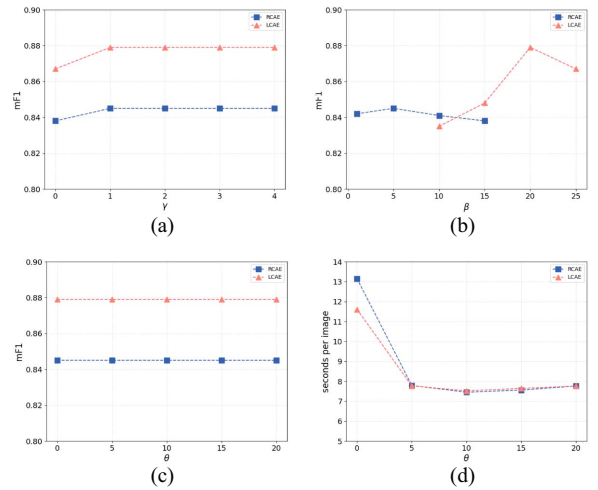


Fig. 13. Sensitivity analysis on the in-house datasets regarding the value of three hyperparameters of the TS-Seg algorithm. (a) Effect of  $\gamma$  on mean F1 value. (b) Effect of  $\beta$  on mean F1 value. (c) Effect of  $\theta$  on mean F1 value. (d) Effect of  $\theta$  on inference speed. (“\$” denotes the greek small letters.)

#### E. Ablation Study of Joint Framework

Given that the performance of the TS-Seg algorithm largely depends on the accuracy of keypoint prediction of the LEED network, we conduct an ablation study to evaluate this impact. The qualitative results are shown in Fig. 15. Specifically, we select one vascular branch as the segmentation target and move one of its endpoints while fixing the other one for comparison.

In Fig. 15(a), we take RCA as an example and fix its ending point. The first and second rows illustrate segmentation results when perturbing the x- and y-coordinate of the starting point, respectively. As shown, when changing the x-coordinate, the performance in terms of F1 score degrades proportionally with increasing APD as the point approximately moves along the vessel centerline. In contrast, when changing the y-coordinate, the F1 scores remain stable when the displaced point lies within

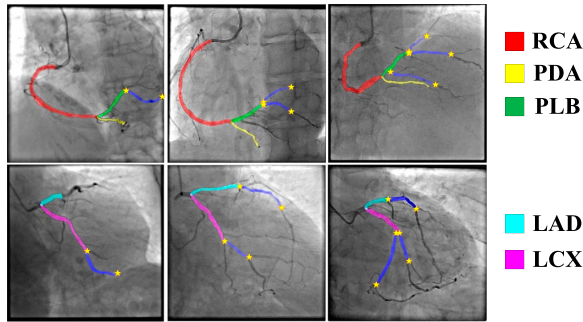


Fig. 14. Examples of the segmentation results of the variant vessel branches. The variant vessels are marked in blue, and the added endpoints are marked with yellow stars. The cases of one, two, and three variant vessels are shown from left to right.

or near the vessel region, due to the constraints of the designed contrastive selection and SVP algorithms. However, large deviations from the vessel region can result in notable performance drops. In Fig. 15(b), we select LAD as an example and fix its starting point. Unlike the RCA, the centerline of the LAD is not approximately parallel to the x- or y-axis. As a result, perturbations to the ending point's x- or y-coordinate have a relatively small influence on the segmentation performance.

These findings indicate that the impact of endpoint deviation on segmentation performance is not linear but depends on the deviation direction and whether the displaced endpoint falls within the vessel region. If the predicted endpoints lie within the vessel, segmentation performance will decline as the deviation of the point position along the vessel centerline increases. When the predicted endpoint deviates from the vessel region by a large distance, it will cause a significant drop in segmentation quality. However, if the deviation is small, the proposed TS-Seg algorithm demonstrates robustness in correcting such deviations, thereby maintaining relatively stable segmentation performance.

## V. DISCUSSION

The main contributions of our work are threefold. First, we introduced a novel paradigm for vascular segmentation that overcame the long-standing challenge of heavy annotation burdens. By employing a two-stage framework, our method achieved vessel semantic segmentation through simple point-level supervision, eliminating the necessity for labor-intensive pixel-wise contour labeling. Second, to accommodate the inherent variability in clinical language, we developed a multi-input-single-output surgical term corpus. This allows for the seamless integration of diverse synonyms, multilingual expressions, or region-specific linguistic habits, thereby ensuring the system's broad generalizability and robustness across different clinical settings. Third, our approach presented superior clinical flexibility: in cases involving anatomical variations, clinicians can manually define endpoints and apply the TS-Seg algorithm alone to achieve accurate analysis of undefined branches. This adaptability enables our framework to be a versatile tool capable of assisting in highly complex interventional scenarios.

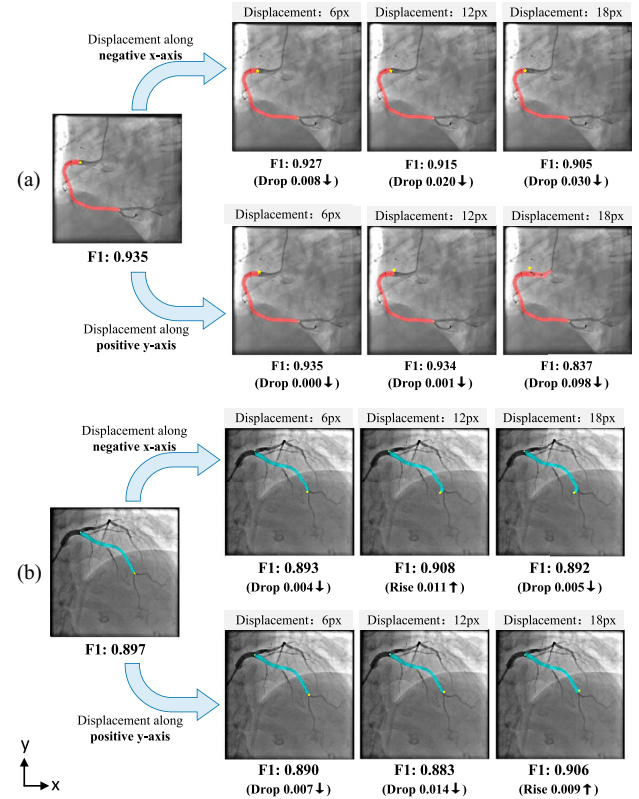


Fig. 15. Qualitative ablation study of the joint framework. Yellow stars and green points denote the moving and fixed endpoints, respectively. The x- or y-axis coordinate of the endpoint is decreased by 6, 12, and 18 pixels in each row. (a) Example of RCA segmentation. (b) Example of LAD segmentation.

Despite its promising performance, our proposed method has several limitations. First, it follows a two-stage framework, where the accuracy of keypoint detection in the first stage heavily influences the segmentation results in the second stage. Thus, this approach requires clinicians to monitor the intermediate results and manually correct keypoint locations if deviations occur. Such manual intervention may affect the efficiency of the workflow. Second, there are many hyperparameters in the settings of the TS-Seg algorithm. If the method is applied to a dataset with large variance, careful parameter tuning is required to achieve satisfactory results. Our future work will focus on developing self-adaptive methods to enhance the model's robustness and generalizability across diverse datasets.

In future studies, it is meaningful to continue exploring the integration of multimodal information for AI-assisted surgeries. The intraoperative communication content of clinicians reflects the surgical progress, which could be utilized to control other machines in the operating room, such as the DSA C-arm, the imaging navigation system, or the surgical robot. This could lead to the development of an intelligent operating room under voice control, improving surgical efficiency and consistency. Furthermore, linguistic information can be leveraged as prior knowledge for selecting the specific vascular branch or surgical instrument according to the clinicians' needs, enabling more adaptive and intelligent assistance.

## VI. CONCLUSION

In this article, we proposed a language-enhanced supervision-reduced framework for precise quantitative analysis of multibranching vessels. By leveraging a flexible two-stage pipeline, integrating automatic endpoint localization with point-guided segmentation, the proposed method effectively mitigates the challenges of inconsistent vessel definitions and heavy annotation burdens. Specifically, the LEED network was developed to enhance keypoint detection accuracy by incorporating clinical linguistic priors through cross-modal feature alignment. Based on the detected endpoints, the devised TS-Seg algorithm could realize semantic vessel segmentation through gray-level features.

Quantitative and qualitative evaluations have been conducted on both in-house and public datasets. The results validated that the LEED network outperformed other keypoint detection benchmarks, benefiting significantly from cross-modal linguistic fusion. Furthermore, the TS-Seg algorithm demonstrated high-quality segmentation with simple endpoint guidance, remaining robust against both LEED errors and hyperparameter fluctuations. To directly enable clinical application, we have generated detailed vessel analysis reports according to segmentation results and medical prior knowledge, which can assist clinicians in both preoperative surgical planning and postoperative outcome evaluation, contributing to more efficient and safer interventional surgeries.

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